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Student Number

**2022** YEAR 12

# Advanced Mathematics

## Task 1 Validation Task

05/12/22

| Q            | Marks      |
|--------------|------------|
| 1            | /2         |
| 2            | /2         |
| 3            | /6         |
| 4            | /2         |
| 5            | /2         |
| 6            | /5         |
| 7            | /2         |
| 8            | /6         |
| 9            | /3         |
| <b>Total</b> | <b>/30</b> |

### General

#### Instructions:

- Reading time – 5 minutes
- Working time – 35 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations
- No white-out may be used

**Total Marks: 30**

***This question paper must not be removed from the examination room.***

*This Validation task constitutes **60%** of the Assessment Task.*

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In Questions 1 – 9, your response should include relevant mathematical reasoning and/or calculations.

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**Question 1 (2 marks)**

If  $\log_{10} 2 = 0.3010$  and  $\log_{10} 3 = 0.4771$ , find:

(a)  $\log_{10} 6$

**1**

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(b)  $\log_{10} \frac{2}{9}$

**1**

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**Question 2 (2 marks)**

Solve the following using change of base. Write your answer to three significant figures.

**2**

$$5^x = 3$$

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**Examination continues on next page**

**Question 3** (6 marks)

Solve for  $x$  if:

(a)  $121^x = 11$

**1**

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(b)  $5^{x+3} = 25$

**1**

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(c)  $\log_2(x + 1) = 3$

**2**

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(d)  $2\log_{10} x + 3 = \log_{10} x^5$

**2**

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**Question 4** (2 marks)

Solve the following for  $x$ :

2

$$81^{x^2} = 3^{-4x-1}.$$

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**Question 5** (2 marks)

Phillip has placed \$2 000 000 in an interest-bearing account. He invested at a rate of 5% p.a. Using the compound interest formula shown below, determine the number of years ( $n$ ) it will take for Phillip to at least double his money.

2

$$A = P(1 + r)^n$$

$A$  (**Amount**) = *Principal plus interest*

$P$  = *Principal*

$r$  = *Interest rate as a percentage*

$n$  = *time in years*

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**Examination continues on next page**

**Question 6** (5 marks)

A logarithmic scale is used to measure the sound level with units in decibels.

$l = 10 \log_{10} \frac{I}{I_o}$ , where  $l$  is in decibels,  $I$  is the intensity of the sound in (watts/m<sup>2</sup>) and  $I_o$  is the threshold of sound intensity detected by the human ear.

- (a) What is the decibel rating of sound at the threshold of hearing (i.e.  $I = I_o$ ). **1**

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- (b) Find the sound level of a conversation, which is 1 million times more intense than the threshold of human hearing. **1**

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- (c) How much greater is the noise intensity of an iPod playing at a peak volume of 115dB ( $I_1$ ) compared with a half volume level of 85 dB ( $I_2$ )? **3**

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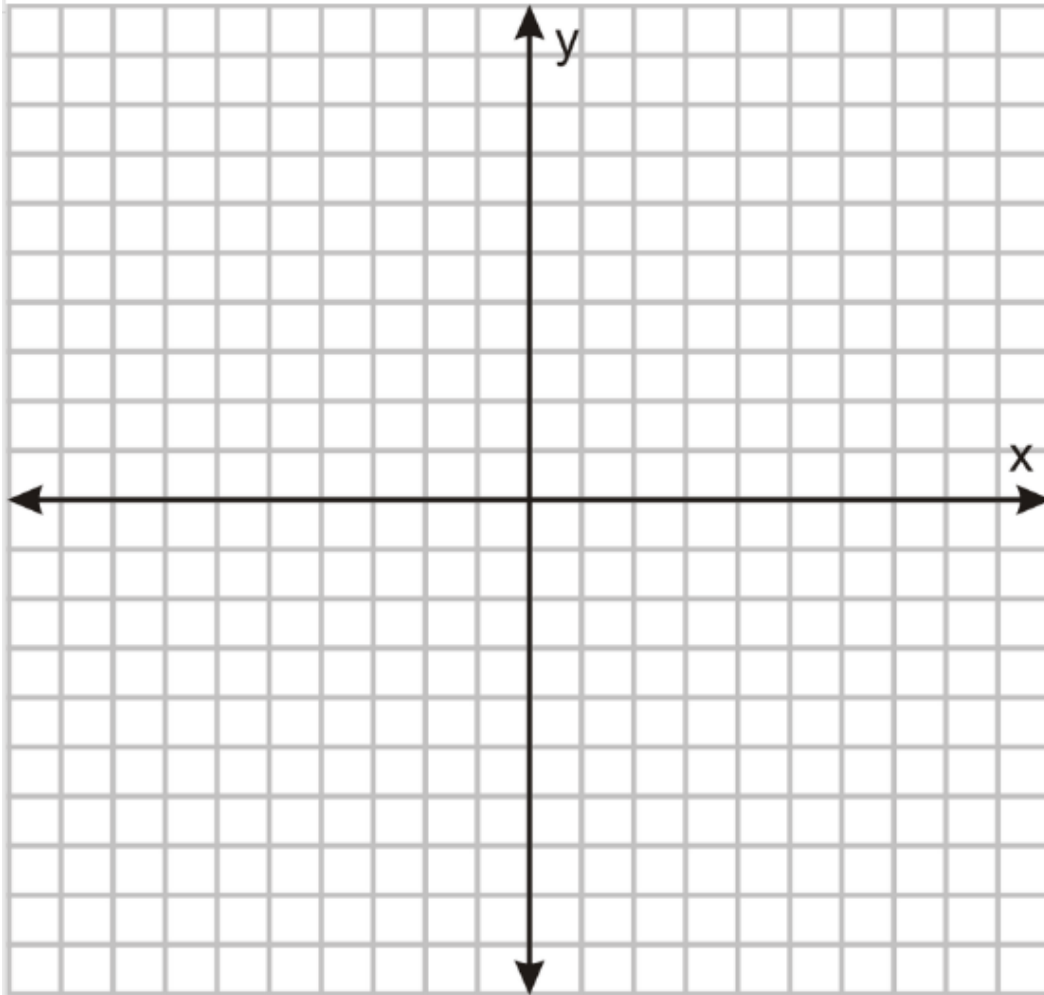
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**Question 7** (2 marks)

Sketch the graph of the following exponential function in the box below. Label any intercepts and the asymptote. 2

$$y = 3e^{-x} + 1$$



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**Examination continues on next page**

**Question 8** (6 marks)

Differentiate the following, simplifying where necessary.

(a)  $y = (x^2 + x + 1)e^{2x}$

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(b)  $y = e^{x - \frac{1}{x}}$

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(c)  $y = \frac{3 + e^x}{e^{4x}}$

2

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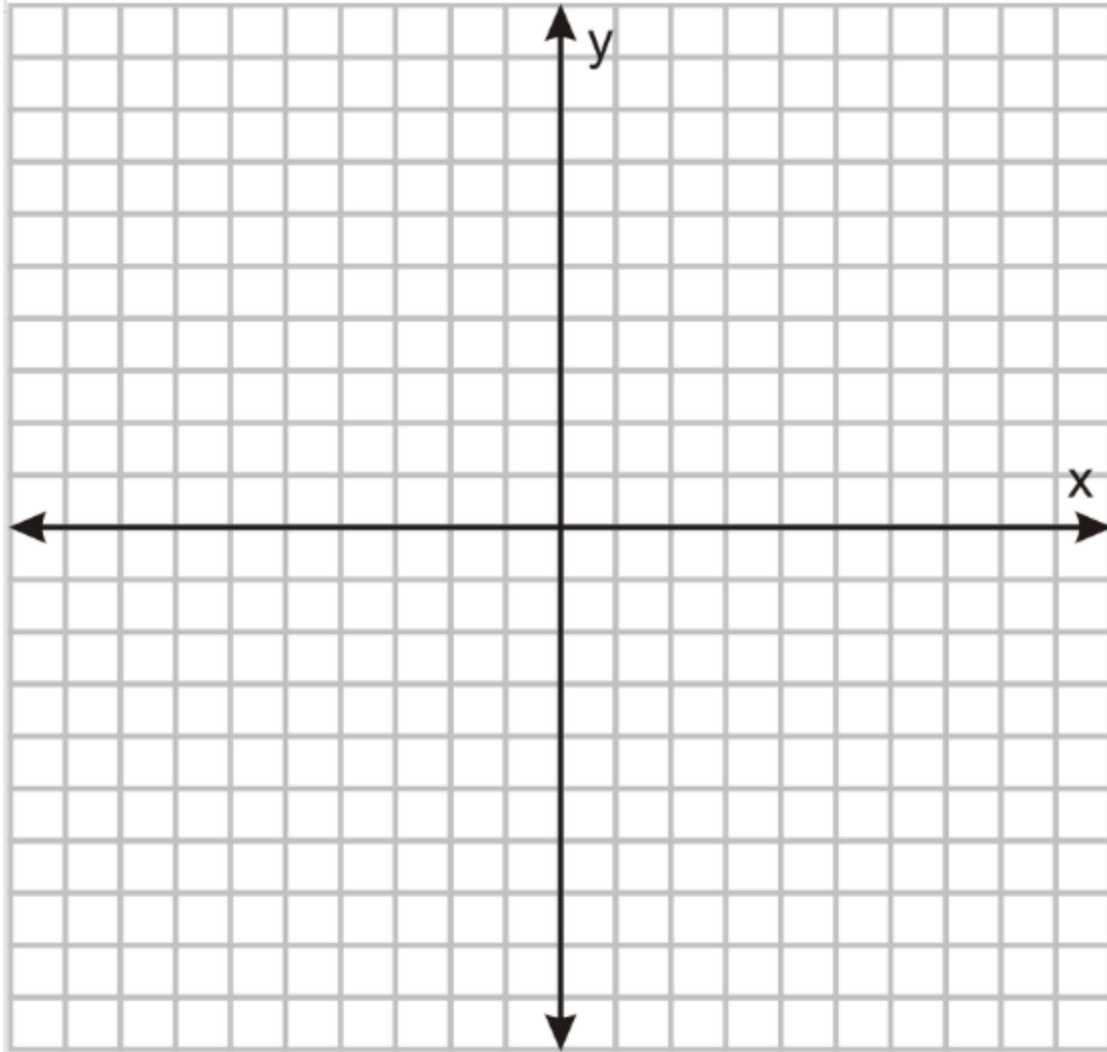
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**Question 9** (3 marks)

Sketch the following graph, clearly indicating the vertical asymptote and labelling all critical points. **3**

$$-y = \log_e(x - 2) + 1$$



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**END OF EXAMINATION**

**Examination continues on next page**

### Extra writing space

**If you use this space, clearly indicate which question you are answering.**

[illegible]

| 1.   | Solutions and Marking Schemes                                                                                                                                                                                                                                                                                                                                                                                                                                            | Markers Feedback                                                     |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1(a) | <p>Question 1 (2 marks)</p> <p>If <math>\log_{10} 2 = 0.3010</math> and <math>\log_{10} 3 = 0.4771</math>, find:</p> <p>(a) <math>\log_{10} 6</math> <span style="float: right;">1</span></p> <p><math>= \log_{10} 2 + \log_{10} 3</math></p> <p><math>= 0.3010 + 0.4771</math></p> <p><math>= 0.7781</math> <span style="float: right;">① - mark for correct answer with working</span></p>                                                                             |                                                                      |
| 1(b) | <p>(b) <math>\log_{10} \frac{2}{9}</math> <span style="float: right;">1</span></p> <p><math>= \log_{10} 2 - \log_{10} 9</math></p> <p><math>= 0.3010 - 2 \times 0.4771</math></p> <p><math>= -0.6532</math> <span style="float: right;">① - mark for correct answer with working</span></p>                                                                                                                                                                              | Some confusion with power rule.                                      |
| 2    | <p>Question 2 (2 marks)</p> <p>Solve the following using change of base. Write your answer to three significant figures. <span style="float: right;">2</span></p> <p><math>5^x = 3</math></p> <p><math>x \ln 5 = \ln 3 \rightarrow</math> This is not using change of base</p> <p><math>x = \frac{\ln 3}{\ln 5} \rightarrow</math> They must show this line for one mark.</p> <p><math>x = 0.683</math> <span style="float: right;">① mark for correct answer</span></p> | Must show change of base in your working, as stated in the question. |

|      |                                                                                                                                                                                                                                                                                                                 |                                                                  |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 3(a) | <p>Solve for <math>x</math> if:</p> <p>(a) <math>121^x = 11</math> <span style="float: right;">1</span></p> $11^{2x} = 11^1$ $2x = 1$ $x = \frac{1}{2}$ <p>or <math>x = \frac{\log_e 11}{\log_e 121}</math></p> <p><math>x = 0.5</math> <span style="float: right;">① - mark for answer.</span></p>             | Some put $x = 2$ . Double check your work.                       |
| 3(b) | <p>(b) <math>5^{x+3} = 25</math> <span style="float: right;">1</span></p> $5^{x+3} = 5^2$ $x+3 = 2$ $x = -1$ <p><span style="float: right;">① - mark for correct answer.</span></p>                                                                                                                             |                                                                  |
| 3(c) | <p>(c) <math>\log_2(x+1) = 3</math> <span style="float: right;">2</span></p> $x+1 = 2^3$ $x+1 = 8$ $x = 7$ <p><span style="float: right;">- ① mark for w/o.</span></p> <p><span style="float: right;">- ① mark for correct answer</span></p> <p><span style="float: right;">(ECF applies.)</span></p>           |                                                                  |
| 3(d) | <p>(d) <math>2\log_{10} x + 3 = \log_{10} x^5</math> <span style="float: right;">2</span></p> $2\log_{10} x + 3 = 5\log_{10} x$ $3 = 3\log_{10} x$ $\log_{10} x = 1$ $x = 10$ <p><span style="float: right;">- ① mark for w/o</span></p> <p><span style="float: right;">- ① mark for correct answer.</span></p> | Some added brackets which changed the question to $2(x+3) = x^5$ |

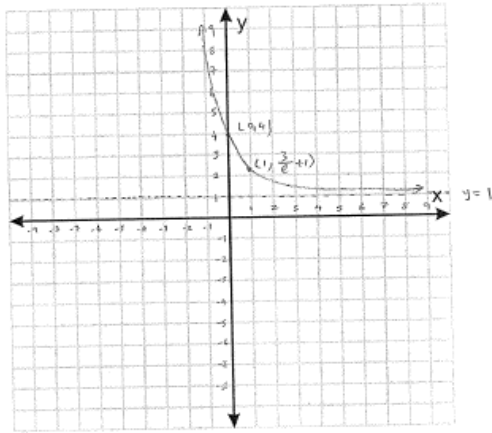
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|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>Solve the following for <math>x</math>: <span style="float: right;">2</span></p> $81x^2 = 3^{-4x-1}$ $3^{4x^2} = 3^{-4x-1}$ $4x^2 = -4x - 1$ $4x^2 + 4x + 1 = 0 \rightarrow \textcircled{1} - \text{mark}$ $(2x+1)^2 = 0$ $\therefore x = -\frac{1}{2} \quad \textcircled{1} - \text{mark for correct answer}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Some were stuck on factorising the non-monic quadratic.                                                                                                                                                                                                                                            |
| 5 | <p>Phillip has placed \$2 000 000 in an interest-bearing account. He invested at a rate of 5% p.a. <span style="float: right;">2</span><br/>Using the compound interest formula shown below, determine the number of years (<math>n</math>) it will take for Phillip to at least double his money.</p> $A = P(1+r)^n$ <p><math>A</math> (Amount) = Principal plus interest<br/> <math>P</math> = Principal<br/> <math>r</math> = Interest rate as a percentage<br/> <math>n</math> = time in years</p> $4\,000\,000 = 2\,000\,000 (1+5\%)^n$ $2 = 1.05^n$ $\log 2 = n \log 1.05$ $n = \frac{\log 2}{\log 1.05}$ $n = 14.20669985 \dots \rightarrow \textcircled{1} \text{ mark for m.f.o.}$ $n = 15 \text{ years} \rightarrow \textcircled{1} \text{ mark for rounding to the next year.}$ | <p>Poorly done.</p> <ul style="list-style-type: none"> <li>- Many assumed that <math>5\% = 5</math></li> <li>- Many did not know how to set up the initial equation</li> <li>- Many did not know to round up to an integer, as interest is only compounded at the <b>end of a year</b>.</li> </ul> |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 6(a) | <p>A logarithmic scale is used to measure the sound level with units in decibels.</p> <p><math>l = 10 \log_{10} \frac{I}{I_0}</math>, where <math>l</math> is in decibels, <math>I</math> is the intensity of the sound in (watts/m<sup>2</sup>) and <math>I_0</math> is the threshold of sound intensity detected by the human ear.</p> <p>(a) What is the decibel rating of sound at the threshold of hearing (i.e. <math>I = I_0</math>). 1</p> <p>at <math>I = I_0</math> <math>10 \log_{10} \frac{I_0}{I_0}</math><br/> <math>= 10 \log_{10} 1</math><br/> <math>= 10 \times 0</math><br/> <math>= 0</math> — (1) mark w/o ...</p>                                                                                                                                                                                                                                                                                                                                                                      |  |
| 6(b) | <p>(b) Find the sound level of a conversation, which is 1 million times more intense than the threshold of human hearing. 1</p> <p><math>I = 10^6 \cdot I_0</math><br/> <math>10 \log_{10} \left( \frac{10^6 \cdot I_0}{I_0} \right) = 2</math><br/> <math>= 60 \text{ dB}</math> (1) mark for substituting<br/> <math>I = I_0</math> and an answer of 60 dB</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| 6(c) | <p>(c) How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (<math>I_1</math>) compared with a half volume level of 85 dB (<math>I_2</math>)? 3</p> <p><math>L_1 = 115 = 10 \log_{10} \left( \frac{I_1}{I_0} \right)</math> <math>L_2 = 85 = 10 \log_{10} \left( \frac{I_2}{I_0} \right)</math><br/> <math>11.5 = \log_{10} \left( \frac{I_1}{I_0} \right)</math> <math>8.5 = \log_{10} \left( \frac{I_2}{I_0} \right)</math><br/> <math>\frac{10}{10} 11.5 = \frac{I_1}{I_0}</math> <math>\frac{10}{10} 8.5 = \frac{I_2}{I_0}</math><br/> <math>I_0 = \frac{I_1}{10^{11.5}}</math> (1) mark <math>I_0 = \frac{I_2}{10^{8.5}}</math> (1) mark<br/> <math>\frac{I_2}{I_1} = \frac{10^{8.5}}{10^{11.5}} = 10^{-3}</math> (1) mark<br/> <math>\frac{I_2}{I_1} = 10^{-3}</math> answer.</p> <p>How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (<math>I_1</math>) compared with a half volume level of 85 dB (<math>I_2</math>)?<br/> L</p> |  |

7

Sketch the graph of the following exponential function in the box below. Label any intercepts and the asymptote. 2

$$y = 3e^{-x} + 1$$



There are no x intercepts.

① - mark for labelling the y intercept

① - mark for a neat sketch

8(a)

Differentiate the following, simplifying where necessary.

(a)  $y = (x^2 + x + 1)e^{2x}$

$$\frac{dy}{dx} = u'v + uv'$$

$$= e^{2x}(2x+1) + (x^2+x+1) \cdot 2e^{2x}$$

$$= (2x^2 + 2x + 2)e^{2x} + (2x^2 + x)e^{2x}$$

$$= e^{2x}(2x^2 + 4x + 3) \quad \text{--- ① mark for simplifying correctly.}$$

u/v.

$$u = x^2 + x + 1$$

$$u' = 2x + 1$$

$$v = e^{2x}$$

$$v' = 2e^{2x}$$

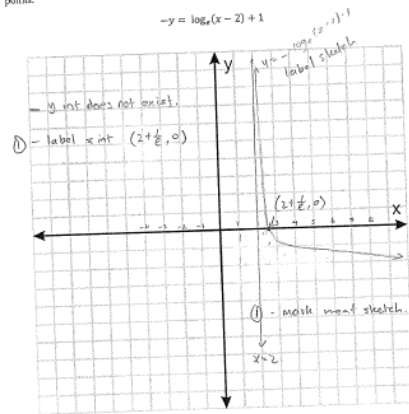
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① - mark for u/v.

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 8(b) | <p>(b) <math>y = e^{x-\frac{1}{x}}</math> let <math>u = x - \frac{1}{x}</math></p> <p><math>\frac{dy}{du} = e^{u-\frac{1}{u}} \cdot (1 + \frac{1}{u^2})</math> <math>\frac{du}{dx} = 1 + \frac{1}{x^2}</math></p> <p><math>\frac{dy}{dx} = (1 + \frac{1}{x^2}) e^{x-\frac{1}{x}}</math> <math>y = e^u</math></p> <p><math>\frac{dy}{dx} = e^u \cdot \frac{du}{dx} = e^{x-\frac{1}{x}} \cdot (1 + \frac{1}{x^2})</math></p> <p>① ans 1</p>                                                                                                                                                                                                                                                                                    | 2 |
| 8(c) | <p>(c) <math>y = \frac{3+e^x}{e^{4x}} \cdot \frac{u}{v}</math> <math>u = 3+e^x</math></p> <p><math>u' = e^x</math></p> <p><math>v = e^{4x}</math></p> <p><math>v' = 4e^{4x}</math></p> <p><math>\frac{dy}{dx} = \frac{vu' - uv'}{v^2}</math></p> <p><math>= \frac{e^{4x} \cdot e^x - (3+e^x) \cdot (4e^{4x})}{(e^{4x})^2}</math></p> <p><math>= \frac{e^{5x} - 12e^{4x} - 4e^{5x}}{e^{8x}}</math></p> <p><math>= \frac{-11e^{5x} - 4e^{5x}}{e^{8x}} = \frac{-15e^{5x}}{e^{8x}} = -15e^{-3x}</math></p> <p>or <math>y = 3e^{-4x} + e^{-3x}</math></p> <p><math>y' = -12e^{-4x} - 3e^{-3x}</math></p> <p><math>= -\frac{12}{e^{4x}} - \frac{3}{e^{3x}} = -8</math></p> <p>① correct numerator</p> <p>① correct denominator</p> | 2 |

9

Sketch the following graph, clearly indicating the vertical asymptote and labelling all critical points.



$$y = -\log_2(x-2) - 1$$